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An identification method of anti-electricity theft load based on long and short-term memory network

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Abstract

Aiming at the problem of many illegal individuals or units adopting various methods to steal electricity under the current power environment, this paper proposes an anti-theft load identification method based on Long Short-Term Memory (LSTM). First, screen users with electricity theft and abnormal behaviors, analyze the collected voltage, current and electric energy by the electricity theft detection equipment, and use the LSTM neural network algorithm to judge the characteristics of the screened suspect users, and initially determine users with abnormal behaviors. Then comprehensively use GPS equipment to send the comparison results of the host to the master station in real time. The master station only needs to be responsible for a small amount of post-processing and statistical analysis, and intelligent diagnosis based on the characteristics of the electricity theft, so that the user can perceive the electricity theft incident on the spot in real time to avoid the loss of electricity bills, and provide a strong theoretical basis for the intelligent operation and maintenance of power companies.

Keywords: load identification, anti-theft, LSTM, Insight;

1. Introduction

In recent years, with the rapid development of the national economy and the continuous improvement of people's living standards, the demand for electricity by residents has increased. In order to save electricity consumption and pursue high profits, many unscrupulous individuals or units have begun to adopt various methods to steal electricity, causing the country and society to suffer huge economic losses. At the same time, with the development of diversification and complexity of power users, the types of power load are also complex and diverse with low

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recognition, which makes it impossible to detect power theft in a timely and accurate manner. This has caused great safety hazards and a lot of economic losses to power companies, and seriously affected their healthy development. Therefore, how to establish an intelligent anti-theft load monitoring system has gradually become the inevitable way for the stable economic development of power grid companies.

Domestic and foreign researchers have conducted a lot of research to realize load monitoring methods and anti-theft technologies. Traditional load identification algorithms need to manually construct features related to electricity theft, which are susceptible to artificial influences, and this has high requirements for system calculations and memory resources at runtime. Some researchers have adopted the technical solution of "terminal collection + master station identification" in the load identification scheme. This scheme enables the master station to bear a huge identification algorithm calculation load after a large number of users access the master station. In addition, from domestic and foreign research, the algorithm has relatively high identification accuracy under a single equipment operating condition, but the stability of the identification accuracy under complex operating conditions varies greatly. Among them, complex operating conditions refer to the fact that useful power, reactive power, harmonics and other characteristic quantities appear close and collide due to the simultaneous operation of multiple devices, which has a high probability of affecting the accuracy of load identification. Therefore, this article uploads the load identification data collected by electricity consumption information to the master station through a dedicated wireless public network, which solves the current communication technology bandwidth and traffic, and does not allow the terminal to upload a large number of intermediate results to the master station. The master station only needs to be responsible for a small amount The post-processing and statistical analysis work of the power consumption information collection system performs intelligent analysis on the collected voltage, current and electric energy, and uploads the identification results to the main station system, reducing the amount of calculation of the main station.

Based on the existing research, this paper uses the electricity consumption information acquisition system to sample the electric energy parameters of the user's total incoming line. First, use the K-means algorithm to cluster and preprocess the user's electricity consumption information to understand the electricity consumption behavior of residential users, and then use the LSTM neural network algorithm to diagnose the user's electricity consumption behavior to achieve accurate perception of the user's electricity consumption type. The LSTM neural network is used to continuously accumulate the characteristics of the acquired power data, and the discrete power theft data in each power consumption information is mapped to a fixed feature vector to provide residents with the power theft data of illegal users, so that the user's power usage rules and priorities energy consumption is clear at a glance, making power consumption more transparent.

2. Power data processing model based on K-means algorithm

The load monitoring system refers to the installation of monitoring equipment at the power entrance, and then by monitoring the user's voltage, current and other signals, the type and operation of a single load in the load cluster can be analyzed. Figure 1 is a structural diagram of the load monitoring system, which collects the total electricity consumption information of the user's electricity load at the power supply entrance.

2.1. Acquisition data processing

Because the user electricity data collected by the system is rather messy, this article first needs to preprocess the electricity data before analyzing the electricity data. The missing data needs to be missing, and the abnormal and excessive data needs to be filtered. When analyzing power data, if the data at a certain time is missing, you can fill in the vacancy based on the data records at the neighboring time, or you can directly ignore the blank data records. In actual operation, this article directly ignores (or deletes) data with null values. Therefore, use standardization to process these power data. It is necessary to analyze the distribution of each attribute data itself to understand the distribution form of each data.

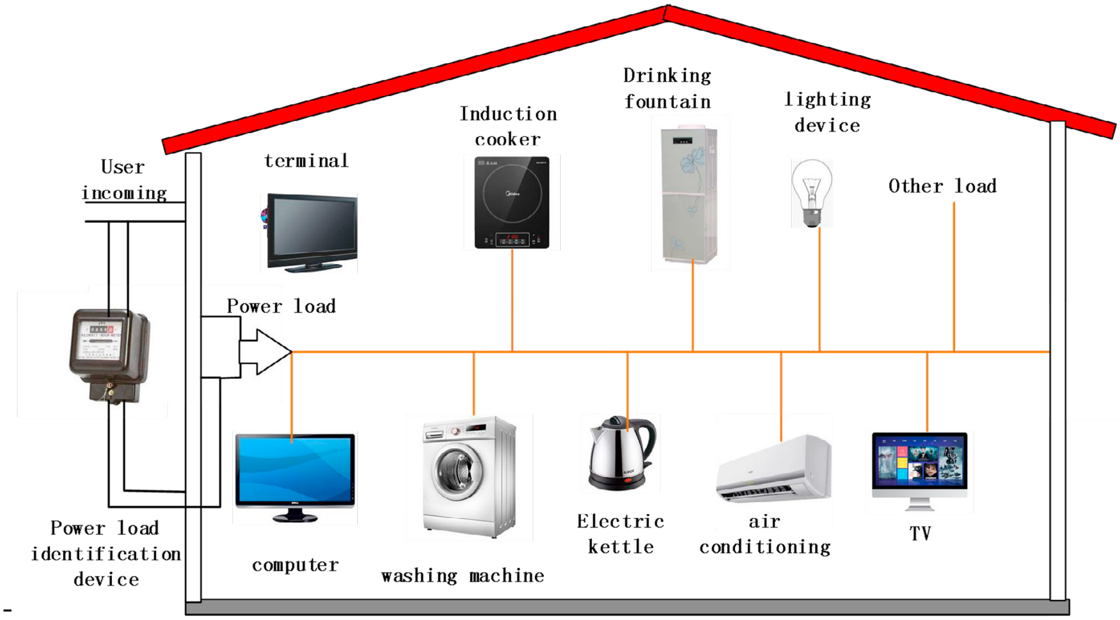

Fig. 1. Load monitoring system structure diagram.

The standardized formula is as follows:

$$y_i = \frac{x_i - \overline{x}}{s} \tag{1}$$

Where $\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(x_i - \overline{x}\right)^2}$, x_i represents data, y_i represents data after normalization, and n represents the number of data.

2.2. Clustering preprocessing based on K-means algorithm

The K-means algorithm is a classic method for preprocessing the electricity consumption information of multiple users. First, the standard electricity theft data is selected as the basis. It is assumed that the user electricity data set is $X = \{x_1, x_2, x_3 \cdots x_n\}$, where n represents the number of users. With the aid of the Euclidean distance formula: $d(x, y) = \sqrt{\left(x_i - y_i\right)^2}$, the data of n users are divided into K categories, and finally the classification results of electricity users stealing electricity are obtained. The clustering preprocessing process of K-means algorithm is shown in the figure below.

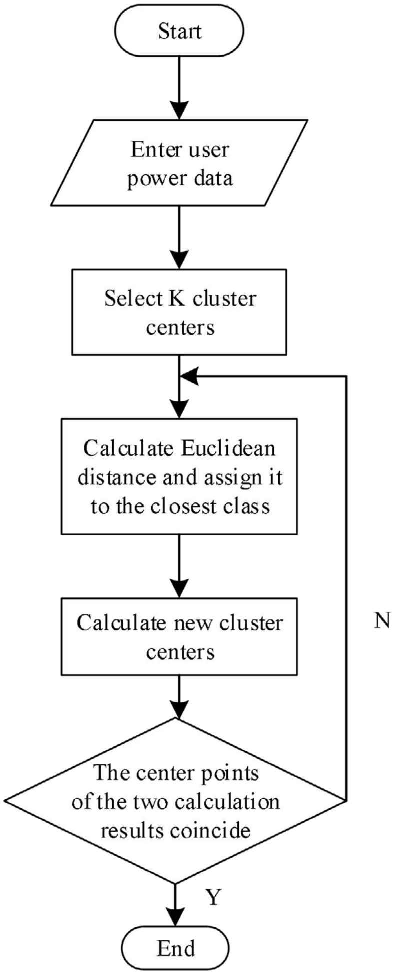

Fig. 2. K-means algorithm flow chart.

3. LSTM-based anti-theft load identification model

The system in this paper can obtain the electricity consumption data of the monitoring terminal equipment at the user's power entrance in real time. By collecting the electricity consumption information of all the users' electric equipment, it can analyze the daily load trend curve of residents' electricity consumption in a short time. Then use LSTM for continuous analysis and processing, deep learning and training, so that the model can identify the power consumption information of each load in the diagnosis system and its operating status. Finally, the load status and operation status of the load cluster are obtained by comparing the electricity consumption information collection system to obtain the electricity stealing status. Figure 3 shows the principle diagram of load identification for anti-tampering.

As shown in Figure 3 above, once the user's electricity consumption data is obtained, the data is preprocessed through the K-means algorithm to reduce dimensionality and noise. Then through the LSTM module to form a feature vector, the feature for training, decision-making. If the feature vector has historical data in the past, the power theft type is diagnosed directly based on the previous load feature. If the extracted feature vector is not in the previous historical data, it needs to be retrained to determine whether the user has theft behavior, so that the learning of the LSTM neural network itself continuously updates the knowledge base. Finally, the system combines GPS and other positioning technologies to lock the location of electricity theft in time, and formulate an anti-theft plan in time.

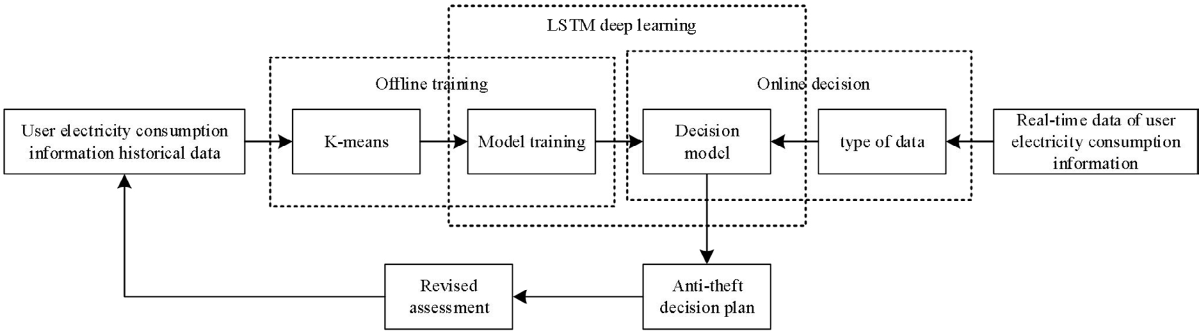

Fig. 3. Schematic diagram of load identification for preventing electricity theft.

3.1. Load characteristics

Load characteristics are important parameters that reflect the power consumption status of users, and reasonable selection of load characteristics is the first step for load identification. Generally, the load characteristics can be divided into the power of the electrical equipment, the load power to the applied electrical equipment, and the use time of the applied electrical equipment. These characteristics ultimately lay the foundation for accurate load identification. The active power P and the reactive power Q are mathematically based on the real-time data of voltage and current sampled in a cycle, and they are defined as follows:

$$P=\frac{1}{T}\sum_{t=0}^{T}V(t)I(t) \tag{2}$$

$$Q=\frac{1}{T}\sum_{t=0}^{T}V\left(t+\frac{1}{4}T\right)I(t) \tag{3}$$

Among them, T represents the number of samples in a waveform period, $V(t)$ represents the voltage of the circuit at time t, and $I(t)$ represents the current corresponding to time t.

The current effective value expression is:

$$I_{rms}=\sqrt{\sum_{i=1}^{\infty}I_i^2} \tag{4}$$

Among them, P is the active power; I_{rms} is the effective value of the current; V and I are the voltage and current of the load respectively; i is the harmonic order.

3.2. Diagnostic process

Assuming that a total of N groups of user electricity consumption data are collected, user information is collected through the electricity consumption information collection system, including user ID, input voltage, current, active power, reactive power, electricity, secondary current, monthly loss value, etc. Categorize these characteristics, and finally get the eigenvalue matrix X of user electricity:

$$X=\begin{bmatrix}X_1\\X_2\\\vdots\\X_n\end{bmatrix}=\begin{bmatrix}ID_1 & t_1 & I_1 & U_1 & P_1 & Q_1 & W_1 & loss_1\\ID_2 & t_2 & I_2 & U_2 & P_2 & Q_2 & W_2 & loss_2\\\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots\\ID_n & t_n & I_n & U_n & P_n & Q_n & W_n & loss_n\end{bmatrix} \tag{5}$$

Among them, $X_1, X_2 \cdots X_n$ represents the power consumption information of N users, ID represents the user's ID ; t represents the sampling time; I is the input current of the user; U represents the input voltage of the user; P represents the active power; Q represents the reactive power; W represents the power consumption of the user; $loss$ represents the power loss of the user. The feature vector corresponding to the sampled data is input into the LSTM network to obtain the diagnosis result, and then combined with the actual survey result to obtain the final diagnosis result. The process is shown in the figure:

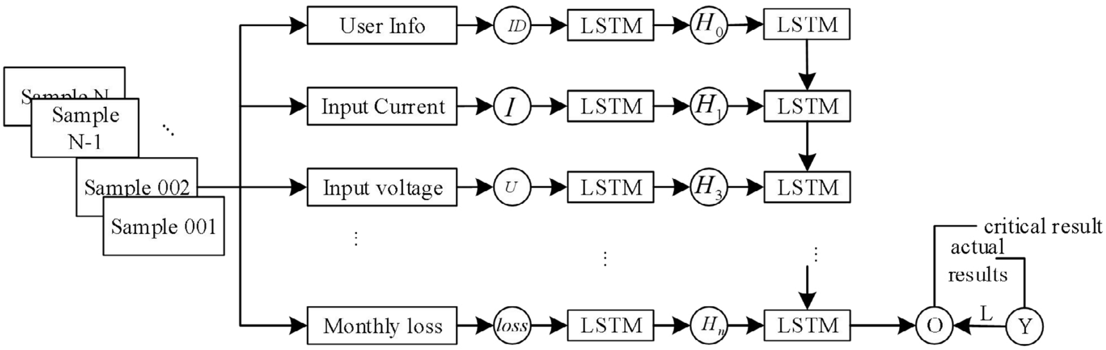

Fig. 4. LSTM-based diagnosis flowchart.

Due to the difference in specific uses and usage time, each user will always cause significant changes in electrical characteristics such as voltage, current, power, and harmonics when the bus is started, stopped, and running. There are obvious differences in electrical characteristics. Therefore, the identification of anti-electricity theft load first requires the establishment of a load characteristic quantity model library of various electric-theft techniques and features, and then uses characteristic quantity extraction and pattern recognition to distinguish the types of electricity consumption of each user, and finally combines actual surveys to obtain the final type of electricity theft judgment.

4. Distributed realization of identification algorithm

In this paper, the anti-theft slave is directly installed on the 10kV line, and the electricity is directly taken through the current transformer, which is simple to install and safe to take. The 10kV line current is quickly sampled and sent to the data through the LORA method. The anti-theft host adopts a modular design, which is installed in the metering box in a guide rail type. The meter data is collected through RS485, and data communication is carried out with the slave through LORA. The host computer compares the primary current data with the meter current data collected by RS485, and can discover metering abnormalities in real time. The remote communication module of the host can communicate with the software of the host station through 4G, and send the comparison result of the host to the host station in real time, so that the user can perceive the electricity theft incident on the spot in real time and avoid the loss of electricity bill. The system structure diagram is shown in Figure 5 below:

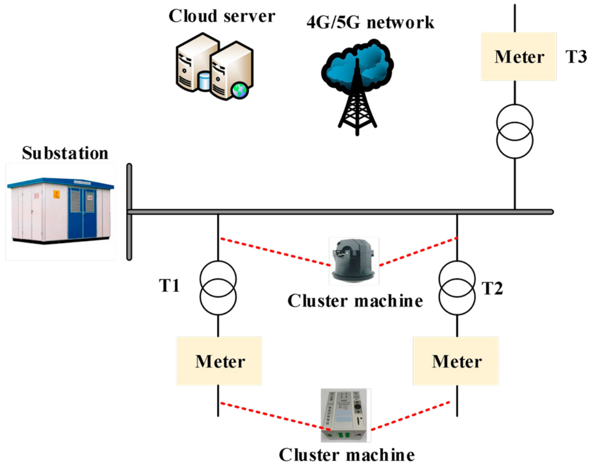

Fig. 5. System structure diagram.

The traditional load identification data is uploaded to the master station through a dedicated wireless public network, which is limited by the bandwidth and traffic of the current communication technology. The terminal is not allowed to upload a large number of intermediate results to the master station. The master station is only responsible for a small amount of post-processing and statistical analysis. Therefore, this article analyzes the collected voltage, current and voltage through smart miniature circuit breakers, and uploads the identification results to the master station system, which reduces the amount of calculation for the master station. Through the analysis and utilization of the user energy efficiency improvement model, the non-linear optimization model is used for strategy design.

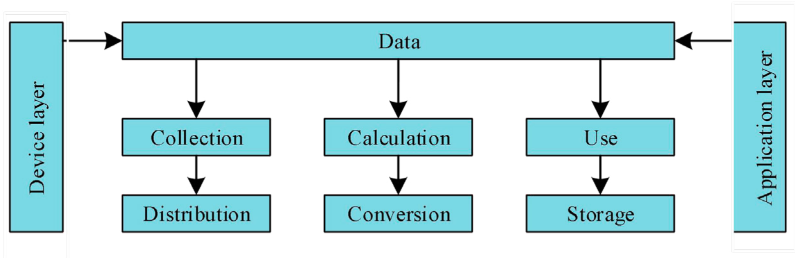

Fig. 6. Distributed system architecture.

Combining the "Big Cloud and Mobility" high-tech and demand analysis to develop energy data service measurement management software for intelligent residential power supply service management. Including power grid repair and accurate fault diagnosis services, electricity safety diagnosis services, guiding users to use electricity reasonably and effectively, etc., to comprehensively improve the management level.

5. Conclusion

Aiming at the problem of illegal electricity stealing encountered by residents, this paper proposes an LSTM-based load identification method for preventing electricity stealing. The system first uses K-means to study in detail the electricity consumption information collection system and the characteristics of electricity theft, and establishes a load identification model based on LSTM. Through continuous training and updating of the network, it identifies illegal electricity theft users. The deep learning method based on LSTM can continuously train the user load characteristics input into the anti-theft load identification system, continuously improve the accuracy of its own model through learning, and then use the trained model to determine the type of power theft. In addition, the anti-theft slave system is used to identify the charge data collected by the user, and the identification results are uploaded to the anti-theft host system, reducing the amount of calculation of the master station system.

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